Sriteja Valaboju

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SUMMARY

Engineer with **4+ years** building and shipping production AI agent systems at Airbus, multi-agent GenAI chatbots, RAG pipelines, and enterprise API integrations. Hands-on across the LLM landscape and agentic tooling with direct experience in prompt design, workflow orchestration, human-in-the-loop checkpoints, RAG/vector retrieval, and enterprise system integration.

EXPERIENCE

Software Engineer - AI — Airbus Group India Pvt Ltd. *April 2025 – Present*

- Productionized **16 AI agents** in Augmented BMS, an app that automates various BMS processes, with evals and observability in place.
- **Architected and delivered Augmented BMS**, a GenAI-powered conversational agent over a corpus of **nearly 4,000** interconnected business documents, serving **80,000 users** with **5,000+ monthly active users** across Airbus; applied ReAct (Reason + Act) patterns for multi-step query resolution.
- Engineered multi-agent workflows using Google ADK for Advanced Search (Deep Mode) and link-analysis features, incorporating human-in-the-loop (HITL) checkpoints; built a Fast Mode using direct semantic search, letting users toggle between Fast and Deep Mode.
- Implemented build and accessing pro-code agents with 3-legged OAuth on the Gemini Enterprise app, testing the flow with Agent Gateway and Agent Identity, and building and registering custom agents and skills in Gemini Enterprise.
- Configured and connected enterprise systems across Augmented BMS and AI Portfolio - handling authentication, data mapping, metadata passing, and fallback configuration for integrations with knowledge bases and internal enterprise tools; troubleshot integration and data-flow issues in production.
- **Designed business and technical architecture** for the AI Portfolio POC/MVP end-to-end and **demoed it to stakeholders** and customers across multiple Airbus functions; leading productionization using React, FastAPI, pgvector/Cloud SQL, and Cloud Run with CI/CD, RBAC, and ReAct-based agent workflows.
- **Architected Airbus**'s **innersourced central Skill Registry** - a categorized library of agent, productivity, and platform skills (e.g., Claude Code skills) with reusable scripts and creation templates, and built a Jenkins CI/CD pipeline to publish and version skills to Artifactory, GCP Artifact Registry, and as Python packages.
- Reduced advanced search latency by **75% (2 min → 30 sec)** through query optimization and efficient vector retrieval pipelines on Vertex AI (LangChain-based RAG); applied prompt engineering to cut token usage and improve output accuracy.
- Cut Docker image sizes by **50% (data ingestion), 80% (frontend), and 10% (backend)**; optimized chunking strategy (**33K → 25K chunks**), reducing Advanced Search query response time **by 60%**.
- Implemented **hallucination detection and response grounding** pipelines in Augmented BMS, flagging factually incorrect or unsupported outputs and anchoring responses to source documents using RAGAS-based evaluation against golden datasets.
- Built usage insights and adoption dashboards for Augmented BMS (Google Data Studio) and AI Forge (React/Recharts), reporting weekly metrics to stakeholders across Airbus functions.
- Built full GCP backend infrastructure using Cloud Run, Cloud Functions, Pub/Sub triggers, Cloud SQL (PostgreSQL), Cloud Logging, and Terraform for IaC, enabling scalable, observable GenAI workloads.
- Refined LLM inference quality by tuning generation parameters (top-k, top-p, temperature) and post-processing pipelines to structure responses and embed contextual document reference hyperlinks inline within outputs.
- Established testing strategy with Jest and Cypress within Jenkins CI/CD; mentored new team members on component architecture and GenAI integration best practices.

Associate Software Engineer — Airbus Group India Pvt Ltd. *August 2022 – March 2025*

- **Owned and led Vue.js frontend** for ModelOps Platform (**official Airbus AI Platform**), architecting Role-Based Access Control and unifying Portal and Monitoring apps under a shell using Vite, Pinia, Chart.js, and a custom Design System.
- Designed architecture and led frontend for ML model creation, SageMaker deployment, and real-time monitoring with drift detection, alerts, and AWS cost/resource optimization dashboards; deployed infrastructure via CloudFormation and Python CDK.
- Worked with AWS Bedrock on the ModelOps Platform: enabled foundation model access and experimented with Bedrock Studio and Strands Agents to prototype agent workflows.
- Co-developed Vue 3 Design System with **50% ownership**, building **50+ configurable, reusable components** with Storybook documentation aligned to Airbus UI/UX standards.

- **Drove official adoption of Vue.js into Airbus engineering** guidelines, expanding the component ecosystem beyond React and Angular and standardizing development across multiple teams.
- Implemented Jest and Cypress testing within Jenkins CI/CD; bootstrapped frontend clones from the ModelOps codebase to accelerate delivery for projects like GraphOps.

ACHIEVEMENTS & CERTIFICATIONS

- **SPOT Award (November 2025)** — Airbus, for contributions to Augmented BMS development and productionization.
- **SPOT Award (March 2025)** — Airbus, for contributions to ModelOps Platform, the **official Airbus AI Platform**.
- **Professional ML Engineer** — Google Cloud
- Advanced Generative AI: Deep Dive (2024) — Google Cloud

TECHNICAL SKILLS

LLMs & Harnesses: Claude, GPT/ChatGPT, Gemini, GLM, DeepSeek; Claude Code, Codex, Antigravity.

GenAI, NLP & RAG: Gemini Enterprise, Google ADK, LangChain, Gemini Enterprise Agent Platform/Vertex AI, Multi-Agent Systems, Agent Orchestration, Agent Gateway, Agent Identity, ReAct Patterns, Human-in-the-Loop (HITL), Semantic Search, RAG, pgvector, Hallucination Detection, RAGAS, Prompt Engineering, LangGraph, Model Context Protocol, TensorFlow, LLMOps, MLOps, Transformers, AI Evaluation

API & Systems Integration: REST APIs, authentication, data mapping, error handling, metadata passing, fallback configuration, knowledge-base integrations

Languages & Frameworks: Python, FastAPI, SQL, JavaScript, TypeScript

Cloud — GCP: Cloud Run, Cloud Functions, Cloud Build, Pub/Sub, Cloud SQL (PostgreSQL), Cloud Logging, BigQuery, Looker Studio, Terraform, IAM

Cloud — AWS: Bedrock, Bedrock Studio, Lambda, S3, DynamoDB, SageMaker, CDK, CloudFormation, VPC, IAM, ECS

Frontend: Vue.js, React.js, Vite, Zustand, Pinia, Tailwind CSS, Chart.js, Recharts, Storybook

Tools: Git, GitHub, JIRA, Jenkins, Docker, Kubernetes, Cypress, Jest, Vitest, Webpack, Figma, Jupyter, Hugging Face, ComfyUI

EDUCATION

Chaitanya Institute of Technology and Science, Hanamkonda*August 2018 – June 2022*

Bachelor of Technology in Computer Science Engineering (CGPA: 8.07), Warangal, India